Exploring Human Steering Performance on Complex 3D Paths

Prototype Design and Construction for a Human-Centered Evaluation System

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Outline

Introduction

Related Work & Design Inspirations

Design Goals & HCAI Principles

Conceptual Design

Prototype Design Methodology

Prototype Artifacts & Interface Design

User Interaction & Task Flow

Implementation Details

Discussion & Conclusion

Introduction

Background & Motivation

- **Steering Law** extends Fitts' Law to trajectory-based tasks:
  $MT = a + b \cdot \frac{L}{W}$
- Modern systems increasingly involve complex 3D environments:
  VR, simulation training, 3D design, robotic teleoperation
- Existing models focus on 2D or simple straight 3D paths
- Path *curvature* and *complexity* significantly affect performance
- **Human-Centered Machine Learning (HCML)** aims to model and support human behaviour while preserving transparency and user control

Objectives & Contributions

Objectives

1. Port Chen et. al's empirical evaluation of a modified Fitts' law to 3D curves [**chen2025curves**]
2. Design and construct a high-fidelity prototype in Unity3D for 3D curve tracking experiments
3. Enable researchers to configure curve parameters and collect accurate steering data

Key Contributions

- A working Unity3D prototype for toroidal curve tracking
- Parameterised curve generation with 6 coefficients (3 sine + 3 cosine)
- Crosshair-based interaction with time-tracking data export

From Requirements to Design

Assignment 1 Recap

- Identified 5 stakeholder personas and 5 aligned usage scenarios
- Specified 12 functional, 8 non-functional, and 4 data/environmental requirements
- Key findings: need for standardised 3D path representations, configurable experiments, and reproducible data

Design Direction

- Prototype translates these requirements into a concrete, testable system
- Focus on accurate data collection over aesthetic user interaction goals
- Curves must remain consistent and trackable across experimental conditions

Related Work & Design Inspirations

Related Work & Theoretical Background

Steering Law Foundations

- Accot & Zhai (1997): trajectory-based movement model [**accot1997beyond**]
- Extensions for curvature and path complexity
- Limited 3D validation [**liu2011revisiting**]

3D Curve Porting

- Chen & Fels (2025): empirical evaluation for curved paths [**chen2025curves**]
- Our work extends this from 2D to 3D toroidal surfaces

Prototyping in HCI

- Iterative design and early usability testing [**sharp2019interaction**]
- Goal-directed design grounded in personas [**cooper2014aboutface**]

HCAI Principles

- Transparency, explainability, human oversight [**shneiderman2022hcai**]
- Guidelines for Human-AI Interaction [**amershi2019guidelines**]

Design Goals & HCAI Principles

Design Goals

Data Collection Goals

- **Accuracy:** Millisecond-precision timing of curve tracking
- **Reproducibility:** Deterministic curve generation from coefficients
- **Configurability:** Adjustable curve parameters via UI sliders
- **Consistency:** Curves must be trackable and well-formed across all parameter ranges

Usability Goals

- **Learnability:** Minimal instruction needed for participants
- **Low Cognitive Load:** Crosshair-based tracking with clear visual feedback
- **Researcher Control:** Experiment setters configure curves via sliders
- **Data Integrity:** Automatic time logging and export

Human-Centered AI Principles in the Prototype

- **Transparency:** All curve parameters and timing data are fully visible and exportable — no hidden computations
- **Human Oversight:** The researcher maintains full control over experiment configuration; no automated adaptation during trials
- **Explainability:** Curve generation is based on an explicit mathematical formula; results are directly interpretable
- **Reliability:** Deterministic rendering from the same coefficients ensures reproducible experimental conditions
- **Ethical Data Handling:** Trial data is stored locally with no personally identifiable information

Conceptual Design

System Overview

Core Concept

- A toroidal curve is generated on a torus surface in 3D space
- The curve is defined by a circle equation plus a sinusoidal height component with 6 adjustable coefficients
- A **crosshair** tracks along the curve; the user drags to follow the path
- The system records **completion time** for each trial

Two User Roles

- **Experiment Setter:** Uses sliders to generate curves with different parameters
- **Participant:** Tracks the curve by dragging along the toroidal path

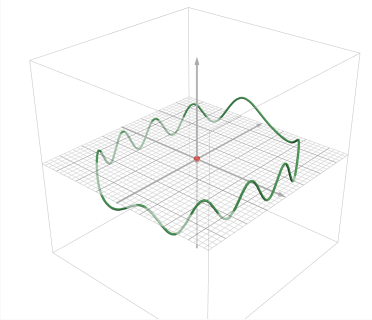


Figure 1: 3D toroidal curve in the prototype environment

Toroidal Curve Equation: Porting from 2D to 3D

Mathematical Foundation

The curve is defined on a toroidal surface by combining a circular cross-section with a sinusoidal height component:

$$x^2 + y^2 = r^2$$

$$z = \sum_{i=1}^{3} \left[\alpha_i \sin(\omega_i \theta) + \beta_i \cos(\omega_i \theta)\right]$$

where $\alpha_1, \alpha_2, \alpha_3$ and $\beta_1, \beta_2, \beta_3$ are the **6 adjustable coefficients** that control curve complexity.

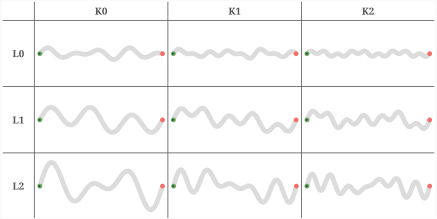


Figure 2: 2D path (Chen et al.)

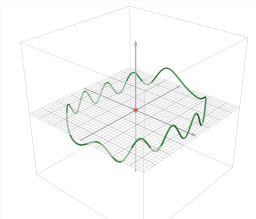


Figure 3: 3D toroidal conversion

Prototype Design Methodology

Design Methodology

Prototyping Approach

1. **Requirements Analysis** (Assignment 1)
   Personas, scenarios, functional requirements
2. **Conceptual Sketching**
   Wireframes for curve display and control UI
3. **High-Fidelity Prototyping**
   Direct implementation in Unity3D
4. **Iterative Refinement**
   Testing curve trackability and data export

Tools & Technologies

- **Engine:** Unity3D (C#)
- **Curve Rendering:** Procedural mesh generation on torus
- **Input:** Mouse-based dragging (desktop)
- **Data Export:** CSV with timestamps
- **Target Platform:** Windows 10+ (desktop)

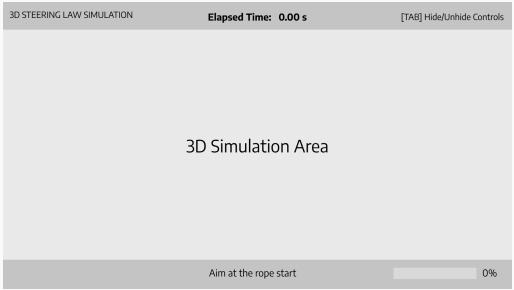
No AI Behaviour Simulation

The prototype is a *data collection tool*; it does not simulate AI responses. AI/ML analysis is a downstream task performed on the exported data.

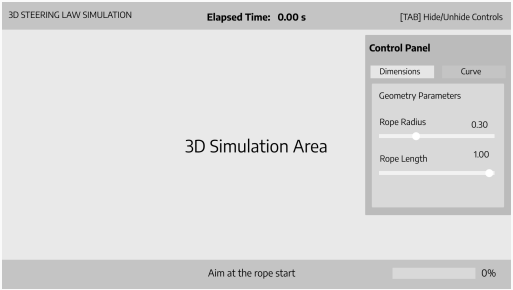
Prototype Artifacts & Interface Design

Wireframes (1/2)

Frame 1



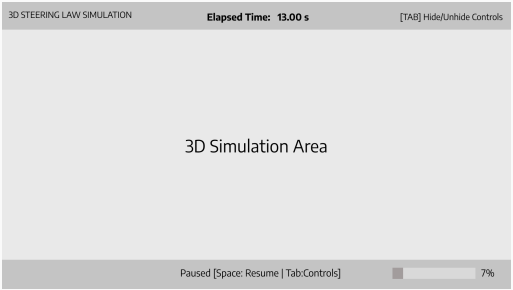
Frame 2



Wireframes (2/2)

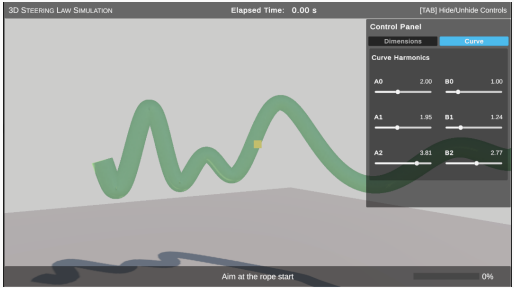
Frame 3

Frame 4

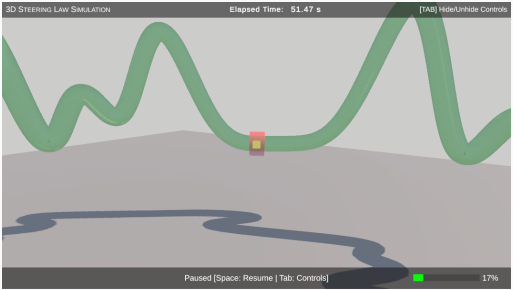


Interface Screens (1/3)

Curve

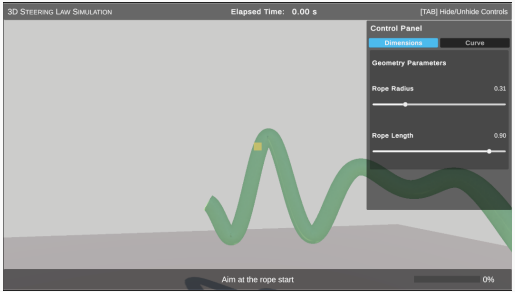


Paused

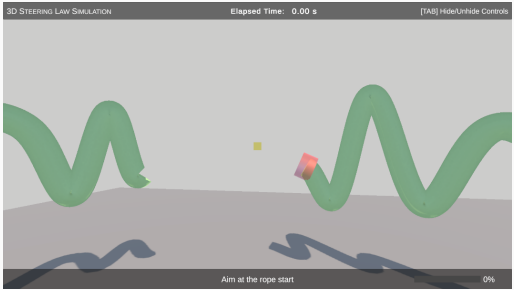


Interface Screens (2/3)

Radius adjustment

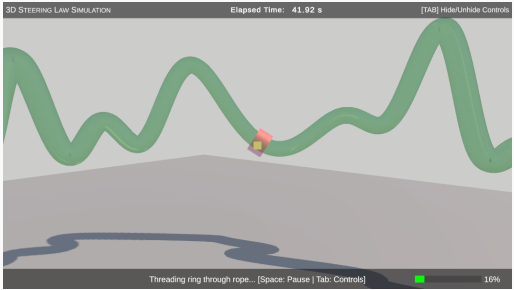


Start of task

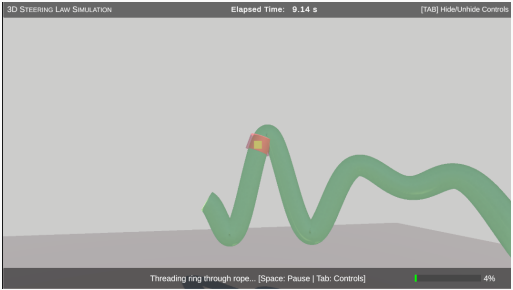


Interface Screens (3/3)

Task in progress



Task in progress with a thicker curve



User Interaction & Task Flow

User Interaction Model

Participant Interaction

1. View the toroidal curve in the 3D viewport
2. Position the crosshair at the start point
3. **Drag along the curve** to track the toroidal path
4. System records completion time automatically
5. Trial ends when the end point is reached

Experiment Setter Interaction

1. Adjust the **6 coefficient sliders** (α_1–α_3, β_1–β_3)
2. Preview the generated toroidal curve in real time
3. Start a new trial for the participant
4. Review exported timing data (CSV)

Input Method

Mouse-based dragging on desktop; VR controller input planned for future work.

Task Flow

Primary Task: *Complete a 3D Toroidal Curve Tracking Trial*

1. **Configure Curve**
   Researcher sets 6 coefficients via sliders
2. **Preview & Validate**
   Visual check that the curve is well-formed and trackable
3. **Start Trial**
   Participant begins dragging the crosshair along the curve
4. **Record Data**
   System captures completion time with millisecond precision
5. **Export Results**
   Timing data saved to CSV for analysis

Data Captured Per Trial

- Curve coefficients (α_1–α_3, β_1–β_3)
- Completion time (ms)
- Trial index and session ID

No AI in the Loop

The prototype operates as a pure data collection instrument. All analysis (e.g., fitting the modified steering law model) is performed offline on the exported data.

Implementation Details

Implementation in Unity3D

Platform & Tools

- **Engine:** Unity3D (2022 LTS or later)
- **Language:** C#
- **Target:** Windows 10+ (desktop)
- **Input:** Mouse (crosshair dragging)

Key Features

- Procedural toroidal curve generation from 6 coefficients
- Real-time curve preview as sliders are adjusted
- Crosshair-based path tracking with visual feedback
- Millisecond-precision timer for trial duration

Architecture

- **CurveGenerator:** Computes toroidal mesh from coefficients
- **CrosshairController:** Handles user input and position tracking
- **TrialManager:** Manages trial lifecycle, timing, and data logging
- **UIManager:** Slider panel, timer display, trial controls

Constraint

VR implementation (Meta Quest) is not yet implemented; currently desktop-only with mouse input.

Discussion & Conclusion

Alignment with Requirements (Assignment 1)

Req. ID	Requirement	Status
FR1	Built in Unity engine	Implemented
FR2	Configurable 3D path geometry	Implemented
FR3	Mouse & VR controller input	Partial (mouse only)
FR4	Millisecond-precision timing	Implemented
FR9	CSV / JSON data export	Implemented
FR11	Meta VR headset support	Not yet
NFR1	$\geq$ 60 FPS on GTX 1650	Met
NFR2	Sub-perceptual input latency	Met

Design Trade-offs & Implications

- **Data Accuracy vs. Visual Richness**
  Prioritised precise timing and consistent curves over elaborate visual design
- **Desktop-First vs. VR-First**
  Chose desktop (mouse) for rapid prototyping; VR integration is planned but not yet implemented
- **Simplicity vs. Feature Completeness**
  Focused on core data collection pipeline; advanced features (practice trials, tutorials, randomisation) deferred
- **Configurability vs. Participant Simplicity**
  Slider controls are exposed only to the experiment setter; participants see only the curve and crosshair

Design philosophy: The prototype is an *instrument for data collection*, not a polished end-user application.

Limitations

Prototype Limitations

- VR mode not yet implemented — currently desktop-only
- No tutorial or practice trial system
- Limited to mouse-based input; no VR controller support
- No boundary violation detection yet
- Trial randomisation not implemented

Methodological Constraints

- Prototype not yet evaluated with real participants
- Curve trackability not empirically validated across all coefficient ranges
- Data analysis (fitting steering law model) is offline and not part of the prototype
- Limited accessibility features

Future Work

- **VR Integration:** Port the prototype to Meta Quest headsets with VR controller input
- **Pilot Study:** Conduct initial user studies to validate curve trackability and data quality
- **Extended Data Logging:** Add trajectory capture, boundary violation detection, and trial randomisation
- **Steering Law Fitting:** Analyse collected data to evaluate the modified Fitts' law in 3D
- **Accessibility:** Add pause/resume, fatigue-aware breaks, and adjustable tolerances
- **ML Analysis Pipeline:** Build downstream tools for outlier detection, movement time prediction, and interpretable models

Conclusion

- Designed and constructed a **high-fidelity Unity3D prototype** for studying human steering performance on complex 3D toroidal curves
- The curve is defined by **6 adjustable coefficients** (3 sine + 3 cosine), enabling systematic variation of path complexity
- Users track the curve via a **crosshair-based dragging interaction**; the system records completion time with millisecond precision
- The prototype is grounded in **HCAI principles**: transparency, human oversight, and reproducibility
- Provides the foundation for **empirical evaluation** of the steering law on 3D curved trajectories

Thank You!

Questions?

References i